

COMPUTER NETWORKS

Week 1 & Week 2 | Tutorial and Examination Questions with Answers

Based on: Tanenbaum & Wetherall — Computer Networks, 6th Edition

Instructor: Dr. Mohammad Shahidul Islam

SECTION A: TUTORIAL QUESTIONS

The following questions are designed for tutorial sessions. Students are encouraged to discuss answers in groups before referring to the model answers provided.

Part 1: Week 1 — Foundations of Computer Networks

Q1. Define a computer network. Why is the term 'autonomous' significant in its definition?

A computer network is a system that connects two or more autonomous computers so they can communicate and share resources such as files, hardware, and software applications.

The term autonomous is significant because it distinguishes a true network from a distributed system or a master-slave configuration. Each computer in a network operates independently — it does not require instructions from another computer to function. This means that the failure or removal of one machine does not necessarily stop the others from operating.

Q2. Differentiate between LAN, MAN, and WAN with one real-world example for each.

Network Type	Coverage Area	Real-World Example
LAN (Local Area Network)	Room, building, or campus (up to 1 km)	University computer laboratory
MAN (Metropolitan Area Network)	A city or town (up to 10 km)	Cable television network across Dhaka city
WAN (Wide Area Network)	Country or entire world (100 km and beyond)	The Internet

The key distinctions are geographical coverage, ownership, and typical speed. A LAN is owned by a single organization and offers the highest speed. A WAN spans large geographical distances and typically uses public infrastructure such as telephone lines, fiber optics, or satellites.

Q3. List and briefly explain four major applications of computer networks.

The four major applications are as follows:

- **Resource Sharing:** Multiple users can share hardware (printers, hard disks) and software installed on a server, eliminating the need for each user to have their own copy or device.
- **Information Sharing:** Files and databases can be accessed from any connected location. A hospital database accessible from multiple departments is a practical example.
- **Communication:** Networks enable e-mail, voice over IP, video conferencing, and instant messaging. Data sent from Bangladesh to the United States via e-mail arrives in seconds.
- **Distributed Processing:** Large computational tasks are divided among many computers. Scientific research, such as climate modelling, uses thousands of networked computers to share the workload.

Q4. Compare Bus, Star, Ring, and Mesh topologies. For each, state one key advantage and one key disadvantage.

Topology	Key Advantage	Key Disadvantage
Bus	Simple and inexpensive to set up	Single cable failure disables the entire network
Star	Failure of one cable affects only one device	Failure of the central hub/switch disables the entire network
Ring	Orderly data transmission; each node gets a turn	A single node failure can disrupt the ring (single ring)
Mesh	Highly reliable; multiple redundant paths exist	Very expensive due to the large number of cables required

Q5. What is a protocol? Explain the terms 'peer entities', 'interface', and 'protocol stack' as used in layered network architecture.

A protocol is a formally defined set of rules governing how two communicating entities exchange data. These rules specify message format, sequencing, error handling, and synchronization.

- **Peer Entities:** Entities (software processes or hardware components) that reside in the same layer on two different machines are called peer entities. They communicate logically using the same protocol. For example, the Network Layer on Host A communicates logically with the Network Layer on Host B.
- **Interface:** An interface is the boundary between two adjacent layers in the same machine. It defines what services the lower layer provides to the upper layer, without exposing internal implementation details.

- **Protocol Stack:** A protocol stack is the complete set of protocols used by a system, with one protocol defined per layer. All layers together constitute the protocol stack. For example, when a user browses a website, the stack is: HTTP (Application), TCP (Transport), IP (Network), Ethernet (Data Link), and the physical medium.

Q6. Explain the four service primitives used in layered network communication, and describe the sequence in which they occur during a connection.

Service primitives are standardized commands that allow adjacent layers to communicate with each other. There are four primitives:

Primitive	Direction	Meaning
REQUEST	Upper layer to lower layer	The upper layer asks the lower layer to perform a service
INDICATION	Lower layer to upper layer	The lower layer informs the upper layer that an event has occurred
RESPONSE	Upper layer to lower layer	The upper layer responds to a prior INDICATION
CONFIRM	Lower layer to upper layer	The lower layer confirms that a REQUEST has been completed

Sequence: A connection begins with a CONNECT.request from the initiating side. The remote layer notifies its upper layer via CONNECT.indication. The remote upper layer replies with CONNECT.response. Finally, the originating lower layer sends CONNECT.confirm, completing the connection setup.

Q7. Describe the seven layers of the OSI Reference Model, stating the primary function and one associated protocol or technology for each layer.

Layer No.	Layer Name	Primary Function	Protocol / Technology
7	Application	User-facing network services; identifies and authenticates communicating parties	HTTP, SMTP, FTP, DNS
6	Presentation	Data translation, encryption, and compression between formats	JPEG, SSL/TLS
5	Session	Establishes, manages, and terminates communication sessions	NetBIOS, PPTP
4	Transport	End-to-end reliable delivery; segmentation and sequencing of data	TCP, UDP
3	Network	Routing packets from source to destination; logical addressing	IP, ICMP; Router
2	Data Link	Framing, error detection, and flow control between adjacent nodes	Ethernet, HDLC; Switch

1	Physical	Transmission of raw bits over the physical medium	Cables, NIC, Hub
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Q8. Explain the HDLC frame structure and identify the three types of HDLC frames and their purposes.

HDLC (High-Level Data Link Control) is a bit-oriented protocol at the Data Link Layer. Every HDLC frame follows this structure:

Frame Field	Description
Flag (01111110)	Marks the start and end of each frame. Byte stuffing inserts a 0-bit after every five consecutive 1-bits to prevent false flag detection in data.
Address Field	Identifies the destination terminal on multi-drop links.
Control Field	Contains sequence numbers, acknowledgements, and frame type indicators.
Data Field	The user payload; variable in length.
CRC (Checksum)	A Cyclic Redundancy Check value used for error detection.
End Flag (01111110)	Mirrors the start flag, signalling the end of the frame.

The three HDLC frame types are:

- Information Frame (I-Frame): Carries actual user data along with sequence numbers and piggybacked acknowledgements. Uses a sliding window protocol allowing up to 7 unacknowledged frames in transit (3-bit sequence number).
- Supervisory Frame (S-Frame): Used for flow and error control. The four sub-types are: RECEIVE READY (RR), REJECT (REJ), RECEIVE NOT READY (RNR), and SELECTIVE REJECT (SREJ).
- Unnumbered Frame (U-Frame): Used for connection control operations such as establishing (SABM), disconnecting (DISC), and reporting frame rejection errors (FRMR).

Part 2: Week 2 — Data Link Protocols and LAN Standards

Q9. What is PPP? Describe its three key features and explain the purpose of LCP and NCP.

PPP (Point-to-Point Protocol) is the standard data link protocol used for direct connections between two devices over point-to-point links, such as a home router connecting to an Internet Service Provider. PPP is based on the HDLC frame format.

Its three key features are:

- Framing Method: PPP defines a precise frame format that includes error detection, ensuring that transmitted data can be verified for correctness at the receiving end.
- LCP (Link Control Protocol): Responsible for opening, configuring, testing, and closing the point-to-point link. LCP manages the entire lifecycle of the PPP connection.
- NCP (Network Control Protocol): Negotiates network-layer-specific options for each supported protocol (for example, one NCP instance for IP, another for IPX, and another for AppleTalk).

Q10. Describe the PPP frame format, explaining the purpose and value of each field.

Field	Size / Value	Purpose
Flag	1 byte = 01111110	Marks the start and end of each PPP frame. Byte stuffing is applied if this pattern appears in the payload.
Address	1 byte = 11111111	Always set to broadcast; accepts the frame at all stations on the link. No individual addressing is needed on a point-to-point link.
Control	1 byte = 00000011	Specifies unnumbered mode. No sequence numbers or acknowledgements are used by default.
Protocol	2 bytes	Identifies the payload type. A leading 0-bit indicates a network layer packet (e.g., IP); a leading 1-bit indicates a control message (LCP or NCP).
Payload	Up to 1500 bytes	Carries the actual data. The maximum size is negotiable; the default is 1500 bytes.
Checksum	2 bytes (default)	Detects errors in the frame. Negotiable up to 4 bytes for enhanced error detection.

Q11. Describe the phases of a PPP connection from establishment to termination.

A PPP connection progresses through the following six phases in sequence:

Phase	Description
Dead	The physical link is inactive. No communication is occurring.
Establish	LCP frames are exchanged to negotiate and set up link parameters (frame size, authentication method, etc.).
Authenticate	Optional phase in which the identities of the communicating parties are verified using a mutually agreed authentication protocol.
Network	NCP negotiates network-layer parameters, such as assigning an IP address to the connecting device.
Open	The link is fully operational and user data is being transferred between the two sides.
Terminate	LCP frames are exchanged to close the link in an orderly manner. The link returns to the Dead phase.

Q12. What is ATM? Describe the ATM cell structure and explain the difference between UNI and NNI cell formats.

ATM (Asynchronous Transfer Mode) is a networking technology that transfers data in small, fixed-size units called cells. Unlike conventional networking (where frames can vary in size), every ATM cell is exactly 53 bytes: a 5-byte header plus a 48-byte payload. The fixed cell size ensures predictable and fast switching through the network.

The ATM header contains the following fields: GFC (4 bits, UNI only), VPI (8 bits in UNI, 12 bits in NNI), VCI (16 bits), PT (3 bits — Payload Type), CLP (1 bit — Cell Loss Priority), and HEC (8 bits — Header Error Control).

Format	Interface	GFC Field	VPI Field	Primary Use
UNI	User device to ATM switch	Present (4 bits) — local flow control	8 bits	Most common; used at the network edge
NNI	ATM switch to ATM switch	Absent — replaced by extended VPI	12 bits	Used within the core network for routing flexibility

Q13. Explain why IEEE 802 standards were necessary and identify five important IEEE 802 subgroups.

Before standardization, every manufacturer designed proprietary networking systems that were incompatible with competing products. This prevented mass production, kept prices high, and limited adoption. The IEEE formed the 802 committee to establish common, open standards, enabling interoperability, economies of scale, and widespread deployment.

IEEE Standard	Responsibility
IEEE 802.3	Ethernet (CSMA/CD) — the dominant wired LAN standard
IEEE 802.4	Token Bus LAN
IEEE 802.5	Token Ring LAN
IEEE 802.11	Wireless LAN (Wi-Fi)
IEEE 802.16	Broadband Wireless (WiMAX)

Q14. Explain the operation of CSMA/CD. Why can collisions still occur even when stations check the medium before transmitting?

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the medium access control method used by Ethernet (IEEE 802.3). It operates as follows:

- Carrier Sense (CSMA): Before transmitting, a station listens to the shared bus. If the bus is idle, the station begins transmission immediately. If the bus is busy, the station waits for a random period and then checks again.
- Collision Detection (CD): Even after checking, two stations may sense the bus as idle at the same instant and begin transmitting simultaneously. Their signals collide on the medium. The CD component detects this collision, after which both stations immediately halt transmission, wait for a random back-off period, and then retry.

Collisions are still possible because of propagation delay. There is a finite time between when one station begins transmitting and when its signal reaches another station at the far end of the bus. During this propagation window, the distant station may incorrectly sense the bus as free and begin its own transmission, causing a collision.

Q15. Compare Token Bus (IEEE 802.4) and Token Ring (IEEE 802.5) in terms of topology, access method, collision handling, and typical use.

Attribute	Token Bus (IEEE 802.4)	Token Ring (IEEE 802.5)
Physical Topology	Bus	Ring
Logical Order	Logical ring over physical bus	Physical ring
Access Method	Token passing	Token passing
Collision Handling	No collisions — only token holder transmits	No collisions — only token holder transmits
Error Recovery	Retransmission request attached to the token	Destination station reports error; sender retransmits
Typical Use	Industrial automation environments	IBM corporate LANs; also FDDI fiber-optic backbones

SECTION B: FINAL EXAMINATION QUESTIONS

The following questions reflect the style and standard of final examination questions. Answers are concise, precise, and structured for maximum marks.

Part A: Short Answer Questions

Q1. Define the following terms as used in computer networks: (a) Autonomous computer, (b) Protocol, (c) Interface, (d) Peer entities.

- (a) Autonomous Computer: A computer that operates and functions independently of any other computer. It does not require another machine to direct or control its operations.
- (b) Protocol: A formally defined and agreed-upon set of rules that governs communication between two peer entities, specifying message format, sequencing, error handling, and synchronization.
- (c) Interface: The boundary between two adjacent layers within the same host machine. It defines the services that the lower layer provides to the upper layer without revealing how those services are implemented.
- (d) Peer Entities: Software or hardware entities in the same layer on two different host machines. They communicate logically with each other using the same protocol for that layer.

Q2. State the OSI model memory mnemonic (top to bottom and bottom to top) and list the data unit name for each layer.

Top to bottom mnemonic: All People Seem To Need Data Processing (Application, Presentation, Session, Transport, Network, Data Link, Physical).

Bottom to top mnemonic: Please Do Not Throw Sausage Pizza Away (Physical, Data Link, Network, Transport, Session, Presentation, Application).

Layer	Name	Data Unit
7	Application	Message / Data
6	Presentation	Data
5	Session	Data
4	Transport	Segment (TPDU)
3	Network	Packet
2	Data Link	Frame
1	Physical	Bit

Q3. What are the key differences between the OSI model and the TCP/IP model?

Attribute	OSI Model	TCP/IP Model
Number of Layers	7 layers	4 layers
Primary Purpose	Theoretical reference model for learning and standardization	Practical model actually used on the Internet
Separate Layers	Presentation and Session are distinct layers (6 and 5)	Presentation and Session functions are merged into Application
Lower Layers	Data Link and Physical are separate	Combined into a single Network Access (Host-to-Network) layer
Protocols	Defines the model; protocols came later	Defined around existing protocols (TCP, IP)

Q4. A station using HDLC has a 3-bit sequence number. What is the maximum number of unacknowledged I-frames that can be in transit at any time, and why?

With a 3-bit sequence number, the sequence number space ranges from 0 to 7, giving 8 possible values. Under the HDLC sliding window protocol with Go-Back-N or Selective Reject, the maximum number of outstanding (unacknowledged) I-frames is $2^3 - 1 = 7$ frames.

The window size is limited to one less than the sequence number space to prevent ambiguity. If the sender were allowed to have all 8 frames outstanding simultaneously, the receiver would be unable to distinguish between a new frame and a retransmitted frame in the event of a lost acknowledgement.

Q5. What is byte stuffing in HDLC and PPP? Why is it necessary?

Byte stuffing (also called bit stuffing) is a technique used to prevent the HDLC/PPP flag pattern (01111110) from appearing accidentally within the data payload of a frame.

In HDLC, after every sequence of five consecutive 1-bits in the data field, the transmitter automatically inserts (stuffs) a 0-bit. The receiver, upon detecting five consecutive 1-bits, removes the following 0-bit to restore the original data. This ensures that the flag pattern 01111110 can only appear as a genuine frame delimiter and never as part of user data, maintaining correct frame boundary detection.

Q6. State the exact binary values of the Flag, Address, and Control fields in a PPP frame, and give the meaning of each value.

Field	Binary Value	Meaning
Flag	01111110	Standard HDLC flag byte; marks the start and end of every PPP frame

Address	11111111	Broadcast address; all stations on the link accept the frame. No individual addressing is needed on a point-to-point link.
Control	00000011	Unnumbered Information (UI) frame indicator; PPP operates in unnumbered mode with no sequence numbers and no acknowledgements by default.

Part B: Long Answer / Essay Questions

Q1. With reference to the OSI model, explain the function of each layer and describe how data flows from the Application Layer of Host A to the Application Layer of Host B across a network. Include the concept of encapsulation.

The OSI (Open Systems Interconnection) model provides a structured, layered framework for network communication. Data flows through all seven layers on the sending host, traverses the network, and is then processed upward through all seven layers on the receiving host.

Layer Functions:

- Application Layer (Layer 7): Provides the interface between the network and user applications. Handles authentication, resource availability checks, and high-level protocols such as HTTP, SMTP, and FTP.
- Presentation Layer (Layer 6): Responsible for data translation between different formats, encryption for security, and compression to reduce data size.
- Session Layer (Layer 5): Establishes, maintains, and terminates sessions (dialogues) between applications. Provides synchronization checkpoints to allow recovery from failures.
- Transport Layer (Layer 4): Ensures reliable, end-to-end delivery by segmenting data, numbering segments, and controlling flow and errors. TCP provides reliability; UDP provides speed without guarantees.
- Network Layer (Layer 3): Routes packets from the source to the destination across multiple networks using logical (IP) addresses. Routers operate at this layer.
- Data Link Layer (Layer 2): Breaks data into frames, adds physical (MAC) addresses, and performs error detection. Switches and bridges operate at this layer. HDLC and Ethernet are Data Link protocols.
- Physical Layer (Layer 1): Transmits raw bits over the physical medium (cables, fiber, wireless), defining electrical, optical, and mechanical characteristics.

Encapsulation and Data Flow:

On Host A, each layer adds a header (and sometimes a trailer) to the data received from the layer above — a process called encapsulation. The Application Layer produces a Message. The Transport Layer adds a TCP/UDP header, creating a Segment. The Network Layer adds an IP header, creating a Packet. The Data Link Layer adds a frame header and trailer, creating a Frame. The Physical Layer converts the frame into raw bits and transmits them.

Intermediate routers process data only up to the Network Layer (they read the IP header to determine routing) and then re-encapsulate the packet for the outgoing link.

On Host B, each layer strips its corresponding header — a process called decapsulation — and passes the remaining data up to the next layer. The Application Layer ultimately receives the original message as sent by Host A.

Q2. Compare and contrast CSMA/CD (IEEE 802.3), Token Bus (IEEE 802.4), and Token Ring (IEEE 802.5) as LAN access methods. Discuss topology, collision handling, performance characteristics, and typical applications.

The three LAN access methods each address the fundamental problem of controlling which station transmits on a shared medium, but they do so with different mechanisms, topologies, and trade-offs.

CSMA/CD — IEEE 802.3 (Ethernet):

CSMA/CD uses a bus topology and a contention-based approach. Stations listen before transmitting (carrier sense) and transmit immediately if the medium is free. If two stations transmit simultaneously, a collision occurs. The collision is detected, both stations stop, wait a random back-off period, and retry. Collisions are not preventable, only detectable and recoverable. Under heavy load, collision frequency increases, reducing effective throughput. Under light load, CSMA/CD is highly efficient because stations transmit without waiting for permission. Ethernet is by far the most widely deployed LAN technology due to its simplicity and cost-effectiveness.

Token Bus — IEEE 802.4:

Token Bus uses a physical bus topology but imposes a logical ring order for token passing. Only the station currently holding the token may transmit. Once transmission is complete, the station passes the token to the next station in the logical sequence. Since only one station transmits at a time, collisions are entirely eliminated. Token Bus provides deterministic access guarantees, making it suitable for industrial automation and real-time control environments where predictable latency is more important than raw speed.

Token Ring — IEEE 802.5:

Token Ring uses a physical ring topology. A free token circulates continuously. A station wishing to transmit waits for the free token, attaches its data along with sender/receiver addresses and CRC, and releases the token-frame onto the ring. The destination station reads the data, and the sender eventually removes its own frame from the ring and releases a new free token. Like Token Bus, collisions are impossible, and every station receives a guaranteed turn. Token Ring was widely used in IBM corporate environments and was also the basis for FDDI (Fiber Distributed Data Interface), a high-speed fiber-optic backbone network.

Attribute	CSMA/CD (802.3)	Token Bus (802.4)	Token Ring (802.5)
Topology	Bus	Bus (logical ring)	Ring
Collision	Detected; back-off and retry	None — token prevents collisions	None — token prevents collisions
Access	Contention-based	Deterministic (token)	Deterministic (token)

Performance under load	Degrades with collisions	Consistent and predictable	Consistent and predictable
Typical Application	Office Ethernet LANs	Industrial automation	Corporate IBM LANs; FDDI backbones

Q3. Explain the ATM technology in detail. Describe the cell structure, the significance of the CLP and HEC fields, and the difference between UNI and NNI. Why was 48 bytes chosen as the payload size?

ATM (Asynchronous Transfer Mode) is a connection-oriented, cell-switching networking technology designed to support voice, video, and data over a unified high-speed network.

Cell Structure: Every ATM cell is exactly 53 bytes long, comprising a 5-byte header and a 48-byte payload. The fixed cell size is the defining characteristic of ATM — it allows switches to process cells in hardware at very high speeds without needing to determine variable frame lengths.

The header contains six fields: GFC (Generic Flow Control, 4 bits, UNI only), VPI (Virtual Path Identifier), VCI (Virtual Channel Identifier), PT (Payload Type, 3 bits), CLP (Cell Loss Priority, 1 bit), and HEC (Header Error Control, 8 bits).

Significance of CLP: The Cell Loss Priority bit indicates the relative importance of a cell. If CLP = 0, the cell is of normal priority and should be preserved even under congestion. If CLP = 1, the cell has low priority and the network is permitted to discard it during periods of congestion. This mechanism provides a quality of service distinction between critical and non-critical traffic.

Significance of HEC: The Header Error Control field contains a CRC code that checks the integrity of the 5-byte header (not the payload). It can correct single-bit errors and detect multi-bit errors. Additionally, HEC enables cell boundary delineation — the receiver uses the HEC pattern to synchronize and identify where each cell begins in the bit stream, without any additional framing overhead.

UNI vs. NNI: UNI (User-Network Interface) is the format used between an end-user device and the first ATM switch. It includes the 4-bit GFC field for local flow control and uses an 8-bit VPI. NNI (Network-Network Interface) is used between ATM switches within the core network. The GFC field is absent and the VPI is extended to 12 bits, providing a larger routing address space within the network infrastructure.

The 48-byte payload: The selection of 48 bytes was the result of a political compromise rather than a purely technical decision. European countries preferred a 32-byte payload (favouring lower end-to-end delay for voice), while North American countries preferred a 64-byte payload (favouring transmission efficiency for data). The committee agreed on 48 bytes as a mathematical average, satisfying neither group entirely but enabling a global standard to proceed.

Part C: Fill in the Blank and True / False

Fill in the Blank — with Answers

Q1. The HDLC flag byte is _____, and if this pattern appears in the data, _____ is applied.

Answer: 01111110; byte stuffing (bit stuffing) is applied — a 0-bit is inserted after every five consecutive 1-bits.

Q2. In the PPP frame, the Address field is always set to _____, meaning _____.

Answer: 11111111; broadcast — all stations on the point-to-point link accept the frame.

Q3. An ATM cell is _____ bytes in total, consisting of a _____ byte header and a _____ byte payload.

Answer: 53; 5; 48.

Q4. In the OSI model, routing and logical addressing are handled by the _____ Layer, which uses the protocol _____.

Answer: Network (Layer 3); IP (Internet Protocol).

Q5. IEEE 802.3 defines _____, IEEE 802.4 defines _____, and IEEE 802.11 defines _____.

Answer: Ethernet (CSMA/CD); Token Bus; Wireless LAN (Wi-Fi).

Q6. In TCP/IP, the Application layer corresponds to OSI layers _____, _____, and _____.

Answer: Application (7), Presentation (6), and Session (5).

True / False — with Justification

Q1. TRUE or FALSE: In a Star topology, a failure of the central hub affects only one device.

FALSE. In a Star topology, all devices connect to the central hub or switch. If the central hub or switch fails, the entire network loses connectivity. Only the failure of an individual cable between a device and the hub affects a single device.

Q2. TRUE or FALSE: PPP uses connection-oriented service with sequence numbers and acknowledgements by default.

FALSE. PPP operates in unnumbered mode by default (Control field = 00000011). This means there are no sequence numbers and no acknowledgements are exchanged at the data link layer. Error detection is provided by the Checksum field, but reliable delivery must be handled by upper-layer protocols such as TCP.

Q3. TRUE or FALSE: In Token Bus and Token Ring, only the station currently holding the token is permitted to transmit, thereby eliminating collisions entirely.

TRUE. In both Token Bus (IEEE 802.4) and Token Ring (IEEE 802.5), the token is a control frame that grants exclusive transmission rights to its holder. Since only one station can hold and transmit at any given moment, collisions are structurally impossible under normal operating conditions.

Q4. TRUE or FALSE: The ATM cell size of 48 bytes for the payload was determined solely by technical requirements for voice transmission.

FALSE. The 48-byte payload was the result of a political compromise between differing national preferences. European countries favoured 32 bytes (lower delay for voice), while North American countries preferred 64 bytes (better data transmission efficiency). The committee agreed on 48 bytes as a compromise, making it a political rather than a purely technical decision.

Q5. TRUE or FALSE: The OSI model is the reference model actually implemented on the Internet.

FALSE. The OSI model is a theoretical reference model used primarily for learning and understanding network architecture. The TCP/IP model, which has four layers (Application, Transport, Internet, and Network Access), is the model actually implemented and used on the Internet.

